Discrete Random Variables & Probability Distributions

Statistics Worksheet · Grade 10–12

Name: ________________________________ Date: ___________________

Learning Objectives

- Identify and interpret discrete random variables and their probability distributions
- Calculate probabilities using a probability distribution table, including inequalities and complements
- Construct a probability distribution table from a real-world scenario such as flipping a coin multiple times

Problems

1. The table below shows the probability distribution for the number of pets owned by students in a class. Fill in the missing probability for X = 2 so that the distribution is valid.

X	0	1	2	3
P(X)	0.20	0.35		0.15

2. Using the grade distribution from NCSU Statistics 101 shown below, write in plain words what the probability notation P(X = 3) represents.

Grade	F	D	C	B	A
X	0	1	2	3	4
P(X)	0.02	0.10	0.20	0.42	0.26

3. Using the NCSU grade distribution table, find the probability that a randomly selected student earns a grade of B or higher.

Grade	F	D	C	B	A
X	0	1	2	3	4
P(X)	0.02	0.10	0.20	0.42	0.26

4. Using the NCSU grade distribution table, find the probability that a randomly selected student earns a grade worse than a C (i.e., a D or an F).

Grade	F	D	C	B	A
X	0	1	2	3	4
P(X)	0.02	0.10	0.20	0.42	0.26

5. Using the NCSU grade distribution table, find the probability that a student earns a grade lower than an A using the complement rule. Show both methods.

Grade	F	D	C	B	A
X	0	1	2	3	4
P(X)	0.02	0.10	0.20	0.42	0.26

6. A fair coin is flipped three times and X is the number of heads obtained. List all possible outcomes and complete the probability distribution table below.

X (# of Heads)	0	1	2	3
P(X)				

7. Using the coin-flip distribution from Problem 6, find the probability of getting at least 2 heads when flipping a coin three times.

$$P(X \geq 2)$$

8. A survey at a local school asked students how many extracurricular activities they participate in. The results are shown below. Determine whether this is a valid probability distribution and explain why or why not.

X	0	1	2	3	4
P(X)	0.05	0.30	0.40	0.20	0.10

9. The number of customers who enter a small coffee shop each hour follows the discrete probability and also find the probability that fewer than 2 customers enter.

X	0	1	2	3	4	5
P(X)	0.05	0.10	0.25	0.30	0.20	0.10

10. A teacher grades tests on a scale of 0 to 4 (whole numbers only). It is known that: P(X = 0) = 0.03, P(X = 1) = 0.07, P(X = 3) = 2 times P(X = 2), and P(X = 4) = 0.28. Find P(X = 2) and P(X = 3), then calculate the probability that a randomly selected student scores at least a 2.

$$P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$$

Discrete Random Variables & Probability Distributions — Answer Key

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Answer Key

1. Answer: P(X = 2) = 0.30

X	0	1	2	3
P(X)	0.20	0.35	0.30	0.15

- All probabilities in a valid distribution must sum to 1.
- $0.20 + 0.35 + P(X=2) + 0.15 = 1$
- $0.70 + P(X=2) = 1$
- $P(X=2) = 1 – 0.70 = 0.30$

2. Answer: The probability that a randomly selected student earns a grade of B.

- X represents the grade point value of the letter grade received.
- X = 3 corresponds to a letter grade of B.
- Therefore $P(X = 3)$ = the probability that a student receives a B.

3. Answer: 0.68

- $P(X \geq 3)$ includes X = 3 and X = 4.
- $P(X = 3) = 0.42$ and $P(X = 4) = 0.26$.
- $P(X \geq 3) = 0.42 + 0.26 = 0.68$.
- About 68% of students earn a B or higher.

4. Answer: 0.12

- Worse than a C means a grade below C, so $X < 2$ (X = 0 or X = 1).
- $P(X = 0) = 0.02$ and $P(X = 1) = 0.10$.
- $P(X < 2) = 0.02 + 0.10 = 0.12$.
- About 12% of students earn a D or F.

5. Answer: 0.74

- Method 1 (direct): $P(X < 4) = P(0) + P(1) + P(2) + P(3) = 0.02 + 0.10 + 0.20 + 0.42 = 0.74$.
- Method 2 (complement): The complement of 'lower than A' is 'getting an A', $P(X=4) = 0.26$.
- $P(X < 4) = 1 – 0.26 = 0.74$.
- Both methods confirm 74% of students earn below an A.

6. Answer: P(0)=1/8, P(1)=3/8, P(2)=3/8, P(3)=1/8

X (# of Heads)	0	1	2	3
P(X)	1/8	3/8	3/8	1/8

- All possible outcomes: TTT, TTH, THT, HTT, HHT, HTH, THH, HHH — 8 equally likely outcomes.
- $X=0$: only TTT → 1 outcome → $P(0)=1/8$.
- $X=1$: TTH, THT, HTT → 3 outcomes → $P(1)=3/8$.
- $X=2$: HHT, HTH, THH → 3 outcomes → $P(2)=3/8$.
- $X=3$: only HHH → 1 outcome → $P(3)=1/8$.
- Check: $1/8 + 3/8 + 3/8 + 1/8 = 8/8 = 1$. ✓

7. Answer: 0.50

- $P(X \geq 2)$ includes $X = 2$ and $X = 3$.
- $P(X = 2) = 3/8$ and $P(X = 3) = 1/8$.
- $P(X \geq 2) = 3/8 + 1/8 = 4/8 = 0.50$.
- There is a 50% chance of getting at least 2 heads.

8. Answer: Valid. All probabilities are between 0 and 1, and they sum to 1.05 — wait, 0.05+0.30+0.40+0.20+0.10 = 1.05. NOT valid.

- Check 1: Each probability must be between 0 and 1. All values satisfy this condition.
- Check 2: All probabilities must sum to exactly 1.
- $0.05 + 0.30 + 0.40 + 0.20 + 0.10 = 1.05 \neq 1$.
- Because the probabilities sum to 1.05, this is NOT a valid probability distribution.

9. Answer: $P(2 \leq X \leq 4) = 0.75$; $P(X < 2) = 0.15$

- Part 1 — $P(2 \leq X \leq 4)$: Add probabilities for $X = 2, 3, 4$.
- $0.25 + 0.30 + 0.20 = 0.75$.
- Part 2 — $P(X < 2)$: Add probabilities for $X = 0$ and $X = 1$.
- $0.05 + 0.10 = 0.15$.
- Verify: 0.75 and 0.15 are both less than 1 and make intuitive sense given the distribution.

10. Answer: $P(X=2) = 0.2067 \approx 0.207$, $P(X=3) = 0.4133 \approx 0.413$; $P(X \geq 2) \approx 0.90$

- Set up the equation: $0.03 + 0.07 + P(2) + P(3) + 0.28 = 1$.
- We know $P(3) = 2 \cdot P(2)$, so substitute: $0.38 + P(2) + 2 \cdot P(2) = 1$.
- $0.38 + 3 \cdot P(2) = 1 \rightarrow 3 \cdot P(2) = 0.62 \rightarrow P(2) = 0.62/3 \approx 0.2067$.
- $P(3) = 2 \times 0.2067 \approx 0.4133$.
- $P(X \geq 2) = P(2) + P(3) + P(4) \approx 0.2067 + 0.4133 + 0.28 = 0.90$.
- About 90% of students score a 2 or higher.